# Genesis of Periodic classification Periodic Table

## • Doberiner Triad :

→ Triad

① Classified elements in group of 3 of similar physical & chemical property &

② In triad, at wt of middle element is arithmatic mean of 1st and 3rd element.

| | | |
|---|---|---|
| Li | Cl 35.5 | Ca 40 |
| 7 | Br 80 | Sr 88 |
| Na | I 127 | Ba 137 |
| 23 | $\frac{127+35.5}{2}$ | $\frac{137+40}{2}$ |
| K | 81.25 | 88.5 |
| 39 | | |

$\frac{7+39}{2} = 23$

• It failed bcoz it was only applicable to few elements.

## Newland law of octave

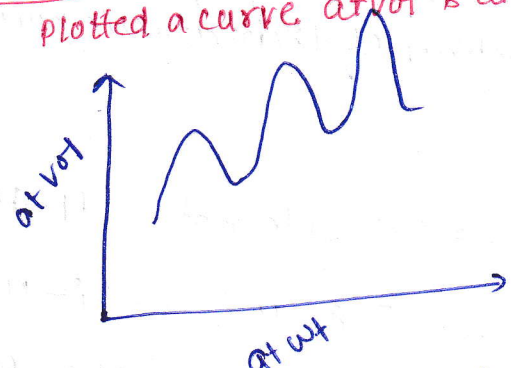
- Elements are arranged in increasing order of at wt.
- Every 8th element had similar property with 1st element. (like musical node)

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 Sa | 2 Re | 3 Ga | 4 Ma | 5 Pa | 6 Dha | 7 Ni |
| | Be | B | C | N | O | F |
| Li | | Al | Si | P | S | Cl |
| Na | Mg | | | | | |
| K | Ca | | | | | |

failed • Valid only up to Calcium.

## Lothermayer

plotted a curve at vol & at wt.




- crest (peak) → alkali
- descending → alkaline
- ascending → halogen
- Trough → Transition elements.

failed : bcoz there was no practical use.

# Mendeleev's Periodic Table

**Mendeleev's Periodic Law** :- properties of elements are periodic function of their atomic weight.

- 9 vertical column & seven horizontal rows.
- Similar nature elements were placed in same vertical groups with increasing at wt.

**Demerit**

# at some places, order of atomic weight is changed to satisfy same physical & chemical property.

| Co | Ni | Te | I |
|---|---|---|---|
| 58.9 | 58.6 | 127.6 | 126.9 |

# Isotopes were not given a separate place.

**merit**

He left gap for elements which were not discovered.

Ga → eka aluminium

Ge → eka silicon

He corrected atomic wt of some elements.

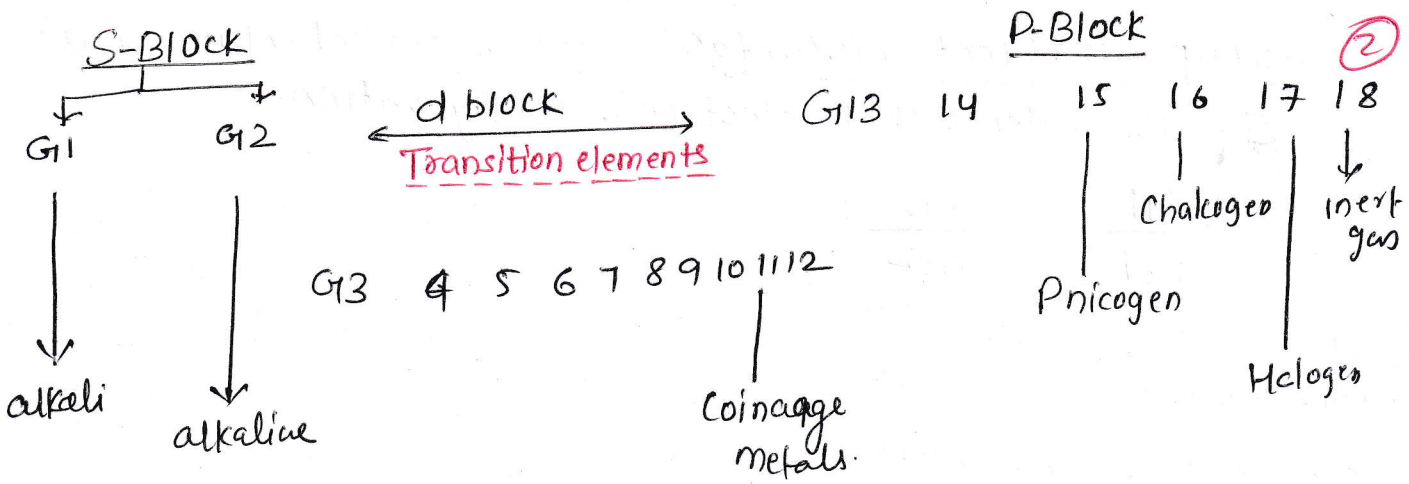
e.g. Be In Au Pt

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## Modern P.T

- Moseley.
- $\sqrt{\nu} \propto Z$
- modern P. Law :- The physical & chemical property of an element are periodic function of their atomic number.
- 7 rows & 18 columns.
- Period

| | | | | |
|---|---|---|---|---|
| 1 | → 1s | → 1 orbital | → 2e⁻ | 2 element → H–He |
| 2 | – 2s 2p | → 4 " | – 8e⁻ | 8 " → Li–Ne |
| 3 | – 3s 3p | → 4 " | – 8e⁻ | 8 " → Na–Ar |
| 4 | – 4s 3d 4p | → 9 " | – 18e⁻ | 18 " → K–Kr |
| 5 | – 5s 5d 5p | → 9 " | – 18e⁻ | 18 " → Rb–Xe |
| 6 | – 6s 4f 5d 6p | → 16 " | – 32e⁻ | 32 " → Cs–Rn |
| 7 | – 7s 5f 6d 7p | – 16 " | – 32e⁻ | 32 " → Fr–Og |




SBlock + PBlock = Representative/main grp element.

f Block (inner Transition)
- Lanthanide Series → Ce(58) Lu(71)
- Actinide Series → Th(90) Lr(103)

## Nomenclature >100

| | |
|---|---|
| 0 nil | 5 Pent |
| 1 un | 6 hex |
| 2 bi | 7 sept |
| 3 tri | 8 oct |
| 4 quad | 9 enn |

101 - Mendelevium
102 - Nobelium
103 - Lawrencium
104 - Rutherfordium
105 - Dubenium
106 - Seaborgium
107 - Bohrium
108 - Hassnium
109 - Meitnerium
110 - Darmstadium
111 - Rontgenium
112 - Copernicium

## Electronic Configrn.

- distribution of electron into orbital is called E.C.
- Period :- The value of n denotes period num.

| | | |
|---|---|---|
| 1st Period | n=1 | $1s^{1-2}$ |
| 2nd | n=2 | $2s^{1-2}$ $2p^{0-6}$ |
| 3rd | n=3 | $3s^{1-2}$ $3p^{0-6}$ |
| 4th | n=4 | $4s^{1-2}$ $3d^{0-10}$ $4p^{0-6}$ |
| 5th | n=5 | $5s^{1-2}$ $4d^{0-10}$ $5p^{0-6}$ |
| 6th | n=6 | $6s^{1-2}$ $4f^{0-14}$ $5d^{0-10}$ $6p^{0-6}$ |
| 7th | n=7 | $7s^{1-2}$ $5f^{0-14}$ $6d^{0-10}$ $7p^{0-6}$ |

Group wise Electronic confgⁿ :- Same vertical column has similar valence shell electronic configuration.

| | G1 | G2 |
|---|---|---|
| | $ns^1$ | $ns^2$ |
| 2P | Li – $2s^1$ | Be – $2s^2$ |
| 3P | Na – $3s^1$ | Mg – $3s^2$ |
| 4P | K – $4s^1$ | Ca – $4s^2$ |
| 5P | Rb – $5s^1$ | Sr – $5s^2$ |
| 6P | Cs – $6s^1$ | Ba – $6s^2$ |
| | | Ra – $7s^2$ |

## S-P-d-f Block

### S Block

last $e^-$ enters in outermost S orbital.
S orbital contains $2e^-$
So it has two groups

G1 $[G]ns^1$ G2 $[G]ns^2$

- Soft, high MP & B.P, reactive, low IE
- +1 & +2 ion form.
- Highly Electropositive & strong Reducing agent
- Metallic character & Reactivity increase down the group.
- ~~Since these are high reactive so never~~

### P-Block

last $e^-$ enters in outermost P orbital.

P orbital – $6e^-$

G13 14 15 16 17 18

GEC = $ns^2 np^{1-6}$

- metal, nonmetal, metalloids
- ~~G18 → full filled confgⁿ → no Reactivity.~~
- ~~G16~~ ~~G17~~ ? ~~Highly -ve electron gain enthalpy.~~
- non-metallic character ~~increased~~ increases from L to R in a Period.
- metallic character increases down the G/P.

### d-block

- last $e^-$ goes into $(n-1)d$ orbital
- also called transition elements (Zn, Cd, Hg)
  G12 exception

d block

| 3d series | 4d series | 5d series | 6d series. |
|---|---|---|---|
| 4th Period | 5th Period | 6th Period | 7th Period |
| Sc–Zn | Y–Cd | La, Hf–Hg | Ac, Rf–Cn |

- GEC → $(n-1)d^{1-10}\ ns^{0-2}$

- ~~none~~ forme colored salt
- Paramagnetic
- Good Catalyst
- Show variable O.state. $Fe^{+2}$ $Fe^{+3}$, $Cu^{+1}$, $Cu^{+2}$

## f block

last $e^-$ → (n-2) f orbital

inner transition element

- Lanthanide (14 elements)
- Actinides (14 elements)

- GEC $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$
- Variable O.state but commonly shows +3 O.state.
- Radioactive
- Transuranium → element after U(92)

## metals, non metals, metalloids

### metal

- Top to bottom metallic character ↑.
- found in left side of P.T
- usually solid (Hg → exception → liq)
- usually high MP & B.P {Ga & Cs → very low MP}
- Good conductor of heat & electricity.

### non metal

- L to R → non metallic character ↑.
- found in righ side of PT
- Usually Solid, gas.
- low MP & B.P (exception B & C)
- Poor Conductor of heat & electricity.

metalloids or semimetal :- Si, Ge, As, Sb, Te, Po, At { both metallic & non metallic property.

## Periodic Trends

### (1) Atomic Radius

Period :- ↓ from L to R (Zeff ↑)

Group :- ↑ from T to B (new shell added)

Zeff & Screening effect

Explain (next page.)

L to R Zeff ↑

T to B almost const/dec

- metallic radius: $r_m = \frac{d}{2}$
- covalent Radii: $r_c = \frac{d}{2}$
- vanderwall radii: $r_A = \frac{d_A}{2}$, $r_B = \frac{d_B}{2}$

V > M > C

# Ionic radius

- Cationic Radius
- Anionic Radius

**Cation Radius $\propto \frac{1}{\text{mag of charge}}$**

**Anion Radi $\propto$ mag of charge**

- Cation is smaller than neutral

$Pb^{+2}$ $Pb^{+4}$

$Mg > Mg^{+} > Mg^{+2}$

$Fe$ $Fe^{+2}$ $Fe^{+3}$ / $Mn > Mn^{+1} > Mn^{+3}$ $Mn^{+3}$ $Mn^{+4}$ $Mn^{+5}$ $Mn^{+6}$ $Mn^{+7}$

- Anion is larger than neutral

$O^{-2} > O^{-} > O$

**A > N > C**

$I^{-} > I > I^{+}$

- **Isoelectronic Species** :- same no of $e^{-}$

$N^{-3}$ $O^{-2}$ $F^{-}$ $Ne$ $Na^{+}$ $Mg^{+2}$ $Al^{+3}$ ($10\ e^{-}$)

$P^{-3}$ $S^{-2}$ $Cl^{-}$ $Ar$ $K^{+}$ $Ca^{+2}$ $Sc^{+3}$ ($18\ e^{-}$)

Size in isoelectronic : A > N > C

- A: $\propto$ mag of charge
- C: $\propto \frac{1}{\text{mag of charge}}$

Que $S^{-2} > Cl^{-} > Ar > K^{+} > Ca^{+2}$

$P^{-3} > S^{-2} > Cl^{-1}$

$N^{-3} > O^{-2} > F^{-} > Ne > Na^{+} > Mg^{+2} > Al^{+3}$

- [redacted]

**Period 2** : $Li < Be > B > C > N > O > F < Ne$

**P3** : $Na > Mg > Al > Si > P > S > Cl < Ar$

**P4** : $Sc > Ti > V > Cr > Mn > Fe = Co = Ni < Cu < Zn$

| G1 | G2 |
|---|---|
| Li | Be |
| ^ | ^ |
| Na | Mg |
| ^ | ^ |
| K | Ca |
| ^ | ^ |
| Rb | Sr |
| ^ | ^ |
| Cs | Ba |

| G13 | Exception |
|---|---|
| B | |
| Ga | d-d contraction → 4th Period → d block |
| ^ | |
| Al | |
| ^ | |
| In | Size is almost same → Lanthanoid contraction → 6th Period f block |
| ^ | |
| Tl | |

**Screening effect** : In a multielectron system valence shell electron experience less attractive force due to replsn blw inner shell electron and valence $e^{-}$. This is called screening / shielding effect.

| G3 | G4 | | | G12 | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
| ^ | ^ | ^ | ^ | ^ | ^ | ^ | ^ | ^ | ^ |
| Y | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd |
| ^ | " | " | " | " | " | " | " | " | " |
| La | Hf | Ta | W | Re | Os | Ir | Pt | Au | Hg |
| ^ | | | | | | | | | |
| Ac | | | | | | | | | |

G13 ——— G18

↓ down the grp size ↑

Que C, N, F, S (size)

S > C > N > F size order

# Ionization energy

energy req to remove an $e^-$ from outermost shell of isolated gaseous atom

$$M_{(g)} + IE \longrightarrow M^+_{(g)} + e^- \quad \text{(endothermic)}$$

$\Delta H$ = +ve always.

## Successive IE

$$M_{(g)} + IE_1 \longrightarrow M^+_{(g)} + e^-$$
$$M^+_{(g)} + IE_2 \longrightarrow M^{+2}_{(g)} + e^-$$
$$M^{+2}_{(g)} + IE_3 \longrightarrow M^{+3}_{(g)} + e^-$$

$IE_3 > IE_2 > IE_1$

↑ Size of $M^{+3}$ is less

Ionization En ∝ oxidation state

## Factors affecting IE

- I.E ∝ Zeff ∝ $\frac{1}{\text{size}}$
- Half filled / full filled are stable. So more IE req to remove electron.
- s > p > d > f

## Trend

Period: L to R : size ↓ IE ↑

Group: T to B : size ↑ IE ↓

P1 H < He

P2 Li Be B C N O F Ne

$2s^1$ $2s^2$ $2p^1$ $2p^2$ $2p^3$ $2p^4$ $2p^5$ $2p^6$

Li < B < Be < C < O < N < F < Ne

P3 Na Mg Al Si P S Cl Ar

Na < Al < Mg < Si < S < P < Cl < Ar

## Exception

B > Tl > Ga > Al > In

C > Si > Ge > Pb > Sn

### In D-Block elements

- G3 → 3d > 4d > 5d
- G 4 5 6 10 → 5d > 4d > 3d
- G 7 8 9 11 12 → 5d > 3d > 4d

Compare $IE_1$ and $IE_2$ of 'O' & 'N'

" $IE_1$ and $IE_2$ of 'Cr' & Mn.

## Electron gain enthalpy

energy released when an $e^-$ is added to an isolated gas atom

$$A_{(g)} + e^- \rightarrow A^-_{(g)} \quad \Delta H = -ve$$

### Successive Eg enthalpy

$$A + e^- \rightarrow A^- \quad \Delta H = -ve \quad EA = +ve$$

$$A^- + e^- \rightarrow A^{-2} \quad \Delta H = +ve \text{ (high value)} \quad (\text{always } +ve) \; EA = -ve$$

$$A + 2e^- \rightarrow A^{-2} \quad \Delta H = +ve \quad EA = -ve$$

### factors affecting Eg enthalpy

- $\Delta H \propto Z_{eff} \propto \frac{1}{size}$
- Hfilled & ffilled are stable so it will not take electron
  so, affinity is less.
  or, Electron gain enthalpy is +ve.

Period → $Z_{eff}$ ↑ EA ↑

size ↑ EA ↓

P1 H > He

P2 Li Be B C N O F Ne

Ne < Be < N < B < Li < C < O < F

P3 Na Mg Al Si P S Cl Ar

Ar < Mg < P < Al < Na < Si < S < Cl

Group:- 3rd Period element has more affinity than 2nd Period element due to more size and less repulsion.

| G13 | G14 | G15 | G16 | G17 |
|---|---|---|---|---|
| Al | Si | P | S | Cl |
| ↓ | ↓ | ↓ | ↓ | ↓ |
| Ga | C | N | Se | F |
| ↓ | ↓ | ↓ | ↓ | ↓ |
| In | Ge | As | Te | Br |
| ↓ | ↓ | ↓ | ↓ | ↓ |
| B | Sn | Sb | O | I |
| | Pb | Bi | | |

G16 < G17 (EA)

Compare EA of F, Cl, O, S.

Cl > F > S > O

# Electronegativity

# Shared pair $e^-$ of covalent bond is attracted by an atom, called EN.

# En → ~~decre~~increase
↓
decrease

En $\propto$ Zeff

En $\propto \frac{1}{\text{Size}}$

En $\propto$ %S

En $\propto$ +ve oxd$^n$ state

$Mn^{+2} < Mn^{+4} < Mn^{+7}$

$O^{-2} < O^{-1} < O < O^{+1} < O^{+2}$

$Fe < Fe^{+2} < Fe^{+3}$

* Pauling scale :- $EN_A - EN_B = 0.1017 \sqrt{E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}}$

* Mullikan scale : $EN = \frac{IE + EA}{2}$

* EN pauling = $\frac{\text{En mulikan}}{2.8}$

* % ionic character = $16 (EN_A - EN_B) + 3.5 (EN_A - EN_B)^2$

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H

C N O F

P S Cl

Br

I

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EA
→ Isolated
→ accept $e^-$

EN
→ Bonded covalent
→ attract $e^-$

→ Acidic nature ↑

↓ Basic Nature ↑

~~BeO~~ . SnO $PbO_2$ ZnO $Al_2O_3$ $Sb_2O_3$ BeO } amphoteric
$SnO_2$ $PbO_2$

CO, $H_2O$, NO, $N_2O$ } neutral oxides

Acidic nature of oxide $\propto$ electronegativity $\propto$ O.St

acidic ↳ $SO_3 > SeO_3 > TeO_3$

$HClO_4 > HBrO_4 > HIO_4$

$HNO_3 > H_2WO_3 > H_3BO_3$

$HClO_4 > HClO_3 > HClO_2 > HClO$

→ acidic nature ↑ as % of oxygen ↑.

$N_2O$
NO
$N_2O_3$
$NO_2$
$N_2O_5$
↓ acidic nature increases

$P_2O_3$
$P_2O_4$
$P_2O_5$
↓ acidic nature increases

MnO
$Mn_2O_3$
$MnO_2$
$MnO_3$
$Mn_2O_6$
↓ acidic nature ↑

MtG . General trend